2017 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2 Hours]

SCC.363 Security and Risk

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 20 marks.

Use a separate answer book for each question.

Question 1

1.a Explain the term *threat model* with an example. **[4 marks]**

1.b Explain the term *security mechanism* with an example. **[4 marks]**

1.c Royal Lancaster Infirmary is considering moving towards an IoT (Internet of Things) infrastructure base where medical devices such as stethoscopes, MRI, and X-ray machines will be managed and operated using smartphone apps installed on staff phones as opposed to the current norm of operating medical devices without using an Internet connection.

Develop a threat model and the corresponding security policy for the IoT hospital.

[12 marks]

Total 20 marks

Question 2

2.a In the context of security, explain the term *user-authentication* with an example.

[4 marks]

2.b Describe a mechanism for non-biometric user authentication with an example.

[6 marks]

2.c Fingerprint-based authentication is commonly proposed as a secure alternative to passwords. However, some security experts disagree. Compare the security of fingerprint vs password-based authentication systems considering such issues as credential generation, recovery from credential loss, theft-resistance, and matters of liability arising from credential theft.

[10 marks]

Total 20 marks

Question 3

3.a Describe four adversary models for traffic analysis with an example for each.

[10 marks]

3.b Develop and describe in detail, a security mechanism based on traffic analysis that can detect distributed Denial-of-Service attacks within the traffic collected on a backbone router (a router that has visibility of all traffic within an enterprise).

[10 marks]

Total 20 marks

Question 4

Malware is a leading attack vector for cybercrime, thus a significant focus of enterprise security defences is to detect malware.

4.a Describe the following terms in the context of malware with an example of each.

i. Command-and-control [2 marks]

ii. Anti-virus [2 marks]

iii. Botnet [2 marks]

iv. Click-malware [2 marks]

v. Rootkit [2 marks]

[10 marks]

4.b Develop a security policy that can prevent the leakage of confidential data from a file server even when the file server is infected with malware seeking to steal data.

[10 marks]

Total 20 marks

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